Supplement S2: The 33 Relations

Arithmetic, Dictionary Choices and Status Brieuc de La Fournière
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S2.0 What this supplement establishes

Earlier versions of this supplement were titled "Complete Derivations" and presented each relation as a theorem with a proof. Version 3.6 withdraws that framing. No geometric map is known that sends the invariants of (K_7, $E_8 \times E_8$) to the Standard-Model parameters, and no rule assigns a formula to an observable (main §3.2). What can be established for each relation is narrower, and this supplement establishes it:

1. **The arithmetic.** Given the declared constants, the closed form evaluates to the stated number. The arithmetic below is exact. Each relation has a Lean statement tying its closed form to its value (`relations_v36_certified`, Arithmon/K7-Lean); the form of each statement (exact rational equality, real closed form, bounds, integer) is given in S3.7.
2. **The dictionary choices.** Which constants the formula uses, which operations beyond integer ratios, and which entries of the residue register it depends on (main §3.3: R1–R12).
3. **The status.** The status of each relation against the most recent data (main §4.1, generated from the dataset) is shown in the last column of each table, with its pull. The column is generated from main §4.1 (`k7_v3_6_s2_status.py`) and is never edited here, so that the status keeps one source; the status vocabulary is defined in main §4.1.

What a relation's arithmetic does not establish: that the formula is the right one for the observable, that the structure requires it, or that the agreement with measurement is evidence for the framework (main §7). Section S2.7 records the alternative formulas found in the corpus for the same observables; their number is a measure of the vocabulary's freedom, not of robustness.

S2.1 Notation

The twenty structural constants and their status (Lie datum, topological, input, combination) are listed in main §3.1. The relations also use the following, each a combination chosen by the framework:

Symbol	Value	Definition	Register
det(g)	65/32	$p_2 + 1/(b_2 + \dim G_2 - N_{gen}) = det_{num}/det_{den}$ (main App. B)	**imposed** normalization (R2); a second expression, Weyl $\cdot\, \alpha_{sum}/2^{Weyl}$, agrees with it on the line $b_2 = 21$ and differs elsewhere
${\kappa_T}^{-1}$	61	$b_3 - \dim(G_2) - p_2$	integer combination without a physical name (R3); a second expression, $b_3 - b_2 +$ Weyl, agrees with it on the line $b_2 = 21$ and differs elsewhere
φ	$(1 + \sqrt{5})/2$	through the McKay correspondence $E_8 \leftrightarrow 2I$	additional step (R9)
1	1	written b_0 in earlier versions	used freely as an additive constant
$2b_2$	42	also $p_2 \times N_{gen} \times \dim(K_7)$	combination
τ	3472/891	$\dim(E_8{\times}E_8) \cdot b_2/(\dim J_3(\mathbb{O}) \cdot H^*)$	used only in Type II entries (S3)

Weyl = 5. Three expressions give 5: $(\dim G_2 + 1)/N_{gen}$, $b_2/N_{gen} - p_2$, $\dim G_2 - \text{rank } E_8 - 1$. Earlier versions called this a "triple identity". It is an arithmetic coincidence among the declared integers; it fixes no further quantity.

$N_{gen} = 3$. The integer identity $(\text{rank } E_8 + N_{gen}) \cdot b_2 = N_{gen} \cdot b_3$, i.e. $11 \cdot 21 = 3 \cdot 77$, is satisfied by $N_{gen} = 3$. It is a consistency check on (21, 77), not the output of an index theorem: the "Atiyah–Singer" expression of earlier versions, $(77 - 8/3 \cdot 21)/7 = 3$, is the same arithmetic written as an index without an operator whose index it is (main §2.4).

S2.2 Gauge sector and Higgs

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A01	$\sin^2\theta_W$ (M_Z, MS-bar)	$b_2/(b_3 + \dim G_2) = 21/91 = \mathbf{3/13} = 0.230769$	scale and scheme (R7)	integer; **proximity**; pull -7.5σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A02	$\alpha_s(M_Z)$	$\sqrt{2}/(\dim G_2 - p_2) = \sqrt{\mathbf{2/12}} = 0.117851$	operation $\sqrt{}$; scale (R7)	algebraic; pull -0.2σ
A03	λ_H	$\sqrt{(\dim G_2 + N_{gen})/\det_{den}} = \sqrt{\mathbf{17/32}} = 0.128847$	operation $\sqrt{}$; det(g) (R2); compared with the tree-level $m_H{}^2/(2v^2)$	algebraic; det(g); pull -1.3σ
A04	$\alpha^{-1}(0)$	$(\dim E_8 + \text{rank } E_8)/2 + H^*/D_{bulk} + \det(g) \cdot \kappa_T = 128 + 9 + 65/1952 = \mathbf{267489/1952} = 137.03330$	det(g) (R2); κ_T (R3); scale $q^2 = 0$ (R7)	det(g), κ_T; **proximity**; pull $\gg 10\sigma$

On A04. The three terms were written as "algebraic", "bulk impedance" and "torsional correction" in earlier versions. None of the three names corresponds to a computation; the correction term is the product of the imposed normalization and the unnamed combination. The relation lies 19.7 ppm below $\alpha^{-1}(0)$, whose uncertainty is 1.5×10^{-10} relative: a proximity (main §4.2).

On A03. A second formula for λ_H is recorded in the corpus (S2.7).

S2.3 Leptons (overdetermined: R8)

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A05	Q_{Koide}	$\dim G_2/b_2 = 14/21 = \mathbf{2/3}$	R8	integer; approximate (R8); pull $+0.4\sigma$
A06	m_τ/m_e	$\dim K_7 + 10 \cdot \dim E_8 + 10 \cdot H^* = 7 + 2480 + 990 = \mathbf{3477}$	R8; the coefficient 10 is not among the twenty constants	integer; approximate (R8); **tension**; pull -2.1σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A07	m_μ/m_e	$\dim(J_3(\mathbb{O}))^\varphi = \mathbf{27}^\varphi = 207.0119$	R8; φ (R9)	φ; approximate (R8); **proximity**; pull $\gg 10\sigma$
A22	m_μ/m_τ	$(b_2 - D_{bulk})/\lvert PSL(2,7)\rvert = 10/168 = \mathbf{5/84} = 0.059524$	R8	integer; approximate (R8); **proximity**; pull $+20.8\sigma$

Four relations bear on two independent mass ratios, and they are mutually incompatible: with 27^φ and 3477, the Koide parameter is 0.666546, not 2/3. None is selected (main §6.1). Earlier versions added that $3477 = 3 \times 19 \times 61$ "factorizes into framework constants"; 19 is not a framework constant, and a factorization is not a derivation. They also cited a "period-integral cross-confirmation" of 3477 from associative 3-cycle volumes, $\ln(m_\tau/m_e) = 8.154$; that number is the logarithm of the measured ratio, used as the target of the volume assignment (S3, entries C17–C19), so the agreement is the target returning, not an independent confirmation. Both statements are withdrawn.

S2.4 Quarks, CKM and boson masses

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A08	m_s/m_d	${p_2}^2 \times \text{Weyl} = 4 \times 5 = \mathbf{20}$	the PDG gives a range (17–22), not a measurement	integer; inside the PDG range
A09	m_c/m_s	$(\dim E_8 - p_2)/b_2 = 246/21 = \mathbf{82/7} = 11.7143$	n/a	integer; pull -1.6σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A10	m_b/m_t	$1/(2b_2) = \mathbf{1/42} = 0.023810$	**mass scheme** (R7): m_b in MS-bar at m_b over the directly measured m_t	integer; **proximity** (in the scheme declared in R7); pull -10.2σ
A11	m_u/m_d	$(1 + \dim E_6)/\lvert PSL(2,7)\rvert = \mathbf{79/168} = 0.470238$	additive 1; the PDG quotes the uncertainty at 90% CL, divided by 1.645 to 1σ (S3)	integer; pull $+0.7\sigma$
A16	$\sin\theta_{12}$ (CKM) $= \lambda$	$\mathrm{fund}(E_7)/\dim E_8 = 56/248 = \mathbf{7/31} = 0.225806$	compared with $\sin\theta_{12}$, not $\sin^2\theta_{12}$; R12	integer; approximate (R12); pull $+0.9\sigma$
A17	A (Wolfenstein)	$(\mathrm{Weyl} + \dim E_6)/H^* = \mathbf{83/99} = 0.838384$	R12	integer; approximate (R12); pull $+0.7\sigma$
A18	$\sin\theta_{23}$ (CKM) $= A\lambda^2$	$\dim K_7/\lvert PSL(2,7)\rvert = 7/168 = \mathbf{1/24} = 0.041667$	compared with $\sin\theta_{23}$, not $\sin^2\theta_{23}$; R12	integer; approximate (R12); pull -0.3σ
A19	m_H/m_t	$\mathrm{fund}(E_7)/b_3 = 56/77 = \mathbf{8/11} = 0.727273$	m_t from direct measurements	integer; pull $+1.8\sigma$
A20	m_H/m_W	$(N_{gen} + \dim E_6)/\dim F_4 = \mathbf{81/52} = 1.557692$	n/a	integer; pull $+0.5\sigma$
A21	m_W/m_Z	$(2b_2 - \mathrm{Weyl})/(2b_2) = \mathbf{37/42} = 0.880952$	n/a	integer; **tension**; pull -2.7σ

The CKM sector is overdetermined (R12). The three relations bear on two Wolfenstein parameters, since $\sin\theta_{23} = A\lambda^2$. With $\lambda = 7/31$ and $A = 83/99$, $A\lambda^2 = 4067/95139 = 0.042748$, while A18 gives $1/24 = 0.041667$: the three relations are 2.6% apart as exact equalities, and at most two can hold. None is selected (main §4.2).

Labels corrected. Earlier versions labelled A16 and A18 "$\sin^2\theta_{12}$(CKM)" and "$\sin^2\theta_{23}$(CKM)" while comparing them with $\lambda = \sin\theta_{12}$ and $A\lambda^2 = \sin\theta_{23}$. The values are sines; the labels are corrected (S3.3).

On the constant 42. Earlier versions presented the appearance of $42 = 2b_2$ in both $m_b/m_t = 1/42$ and $\Omega_{DM}/\Omega_b = (1 + 42)/8$ as "connecting quark physics to large-scale structure". Two formulas that share an integer share a choice of vocabulary; no connection between the quantities is implied, and the sentence is withdrawn.

Dependence noted, not counted. $\sin^2\theta_W$ in MS-bar (A01) and m_W/m_Z (A21) are related in the Standard Model through radiative corrections; the two relations are not independent tests in the strict sense.

S2.5 Neutrino mixing

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A12	$\sin^2\theta_{12}$ (PMNS)	$(1 + N_{gen})/\alpha_{sum} = \mathbf{4/13} = 0.307692$	additive 1	integer; pull -0.2σ
A13	$\sin^2\theta_{23}$ (PMNS)	$(D_{bulk} - \text{Weyl})/D_{bulk} = \mathbf{6/11} = 0.545455$	octant (main §4.3)	integer; **tension** (octant; inside the NuFIT 3σ range, main §4.3); pull $+4.4\sigma$, not Gaussian (main §4.3)
A14	$\sin^2\theta_{13}$ (PMNS)	$D_{bulk}/\dim(E_8\times E_8) = \mathbf{11/496} = 0.022177$	n/a	integer; pull -0.5σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A15	δ_{CP}	dim K_7 · dim G_2 + H* = 98 + 99 = **197**, read as the phase $2\pi \cdot 197/360$	**unit** (R6)	**unit assignment** (R6); pull -0.4σ

The δ_{CP} rule. The prediction is the phase $2\pi \cdot 197/360$, a point value; the unit is a dictionary assignment (R6), and rule D1 of main §11 applies: a measurement whose 3σ interval excludes 197° rejects the relation and the unit assignment together, with no repair clause and no theoretical tolerance added afterwards. The compactification factor 62/69 of earlier versions is addressed in S2.8.

S2.6 Cosmology (overdetermined: R11)

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A23	$\Omega_{\mathrm{DE}} = \Omega_\Lambda$	ln 2 · (b_2 + b_3)/H* = ln 2 · 98/99 = **0.686146**	operation ln 2; R11	ln 2; approximate (R11); pull $+0.2\sigma$
A24	n_s	$\zeta(D_{\mathrm{bulk}})/\zeta(\mathrm{Weyl})$ = ζ**(11)**/ζ**(5)** = 0.964864	operation ζ	ζ values; pull 0.0σ
A25	$\Omega_{\mathrm{DM}}/\Omega_{\mathrm{b}}$	(1 + 2b_2)/rank E_8 = **43/8** = 5.375	additive 1; R11	integer; approximate (R11); pull $+0.1\sigma$
A26	$\Omega_{\mathrm{c}}/\Omega_\Lambda$	$\det_{\mathrm{num}}$/|PSL(2,7)| = **65/168** = 0.386905	R2; R11	integer; det(g); approximate (R11); pull 0.0σ

ID	Relation	Arithmetic	Choices declared	Status (main §4.1)
A27	Ω_Λ/Ω_m	$(\dim G_2 + H^*)/\dim F_4 = \mathbf{113/52} = 2.173077$	R11	integer; approximate (R11); pull 0.0σ
A28	h	$(\lvert PSL(2,7)\rvert - 1)/\dim E_8 = \mathbf{167/248} = 0.673387$	additive 1	integer; pull 0.0σ
A29	Ω_b/Ω_m	$\mathrm{Weyl}/\det_{den} = \mathbf{5/32} = 0.15625$	R2; R11	integer; det(g); approximate (R11); pull -0.1σ
A30	Ω_c/Ω_m	$(\det_{den} - \mathrm{Weyl})/\det_{den} = \mathbf{27/32} = 0.84375$	R2; R11	integer; det(g); approximate (R11); **tension**; pull $+2.5\sigma$
A31	σ_8	$(p_2 + \det_{den})/(2b_2) = 34/42 = \mathbf{17/21} = 0.809524$	R2	integer; det(g); pull -0.3σ
A32	Y_p	$(1 + \dim G_2)/{\kappa_T}^{-1} = \mathbf{15/61} = 0.245902$	additive 1; R3	integer; κ_T; pull $+0.3\sigma$
A33	Ω_m/Ω_Λ	$(1 + \dim J_3(\mathbb{O}))/{\kappa_T}^{-1} = \mathbf{28/61} = 0.459016$	additive 1; R3; R11	integer; κ_T; approximate (R11); pull -0.1σ

The seven density relations bear on two independent densities (flat ΛCDM: $\Omega_b + \Omega_c + \Omega_\Lambda = 1$) and are mutually incompatible as exact equalities (main §4.2):

Quantity	Values implied by the relations
Ω_Λ/Ω_m	113/52 = 2.1731 (A27); 61/28 = 2.1786 (inverse of A33); 2.1862 (A23 with flatness)
Ω_c/Ω_b	43/8 = 5.375 (A25); 27/5 = 5.400 (A30/A29)
Ω_c/Ω_Λ	65/168 = 0.3869 (A26); 0.3873 (A30 × A33); 0.3883 (A30/A27)

A27 and A33 are the same quantity, written with two formulas. The disagreements are below the Planck uncertainties of these ratios, which is why six of the seven pass at 2σ; Ω_c/Ω_m, whose uncertainty is much smaller (0.0019), is in tension at $+2.5\sigma$. The seven cannot hold together; none is selected (R11). Some are redundant and some subsets are compatible: A29 and A30 are equivalent when neutrinos are neglected ($\Omega_m = \Omega_b + \Omega_c$, so that 5/32 + 27/32 = 1), with or without flatness, and the largest compatible subsets have three relations, A29 and A30 with one of A23, A26, A27 or A33 (checked by solving each subset for (Ω_b, Ω_c)). Earlier drafts of this version said that at most two of the seven can be exact; A23, A29 and A30 hold together, and the statement is withdrawn.

Withdrawn. Earlier versions also gave $\Omega_m = \Omega_{DE}/\sqrt{\text{Weyl}} = 0.3068$ (2.7% from Planck) as a "derived" relation, and described ln 2 as a "binary information factor" of the moduli space. The first is an eighth density relation, incompatible with the others; the second names an operation without a computation behind it. Both are withdrawn. The statement that $\Omega_{DM}/\Omega_b = 43/8$ is "exact" (from a dataset value 5.375 ± 0.1 that matched no Planck column) is withdrawn; against the Planck 2018 chains it is 5.364 ± 0.077.

S2.7 Vocabulary freedom: alternative formulas in the corpus

Earlier versions counted, for each relation, the number of distinct expressions in the constants that reduce to the same value (for example 14 for $\sin^2\theta_W = 3/13$, 20 for $Q_{Koide} = 2/3$, 21 for $m_b/m_t = 1/42$), and read a large count as "structural robustness": a value reached by many paths was said to be less likely a coincidence. Version 3.6 reverses that reading (main §7). A value reached by many paths through twenty constants is a value the vocabulary reaches easily; the count measures the vocabulary's freedom (R4, R5). It is not what the look-elsewhere test of main §7 prices: the published values 3/13 and 1/42 lie outside their 2σ windows, and the 30 targets that every competitor reaches are already reached by the grammar's universal leaves, with no constant of K_7 at all (S4.2.2). The counts are kept as such a measure:

Relation	Distinct expressions recorded	Examples
$\sin^2\theta_W = 3/13$	14	N_{gen}/α_{sum}; $b_2/(b_3 + \dim G_2)$; $\dim J_3(\mathbb{O})/(\dim F_4 + \det_{num})$; $(1 + \dim G_2)/\det_{num}$

Relation	Distinct expressions recorded	Examples
$Q_{Koide} = 2/3$	20	p_2/N_{gen}; dim G_2/b_2; dim F_4/dim E_6; rank E_8/(Weyl + dim K_7)
$m_b/m_t = 1/42$	21	$1/(2b_2)$; $(1 + N_{gen})/\lvert PSL(2,7)\rvert$; $p_2/(\dim K_7 + b_3)$; $N_{gen}/(\dim J_3(\mathbb{O}) + H^*)$
$\sin^2\theta_{12}$ (PMNS) = 4/13	28	n/a
$m_H/m_t = 8/11$	19	n/a
$m_u/m_d = 79/168$, $m_H/m_W = 81/52$	1 each	n/a

Second formulas for the same observable. The corpus also contains formulas that give a *different* value for the same observable, chosen at different times:

Observable	Dataset formula	Other formula in the corpus
λ_H	$\sqrt{17/32} = 0.12885$	$b_2(M_1)/(b_3 + b_2(M_2)) = 11/87 = 0.12644$ (TCS building blocks)
Ω_{DE}	$\ln 2 \cdot 98/99 = 0.68615$	$1 - \chi(K3)/b_3 = 53/77 = 0.68831$
m_c/m_s	$246/21 = 11.714$	$b_2(M_1) + \dim K_7/b_2(M_2) = 117/10 = 11.700$
Ω_Λ/Ω_m	$113/52 = 2.1731$	$(\det_{den} - \dim K_7)/D_{bulk} = 25/11 = 2.2727$; $61/28$ (A33 inverted)
θ_{13}	$\sin^2\theta_{13} = 11/496$	$\theta_{13} = \pi/b_2 = 8.571°$; $\theta_{13} = 94/11 = 8.545°$ read in degrees
θ_{23}	$\sin^2\theta_{23} = 6/11$ ($\theta = 47.6°$)	$\theta_{23} = \arcsin(75/99) = 49.25°$ ($\sin^2\theta_{23} = 0.574$)
θ_{12}	$\sin^2\theta_{12} = 4/13$	$\theta_{12} = \arctan\sqrt{((2\pi/25)/(511/884))} = 33.40°$
δ_{CP}	197	$197 \times 62/69 = 177.01$ (S2.8)

The "TCS building-block" formulas use $b_2(M_1) = 11$ and $b_2(M_2) = 10$, Betti numbers of building blocks whose identification is open (main §9). Several alternatives were introduced after a release of the data had moved: the Ω_{DE} alternative was preferred against one Planck release and not the other, and the θ_{13} alternative "in degrees" is a second unit assignment of the kind R6 registers. None of these is part of the 33; they are listed because they show, on the framework's own record, how many formulas the vocabulary offers for one observable and how the choice among them was made.

Patterns. The Fibonacci and Lucas numbers among the constants (2, 3, 5, 8, 13, 21; 7, 11) and the identities among the integers ($\dim G_2 = p_2 \cdot \dim K_7$, $b_2 = N_{gen} \cdot \dim K_7$, $\lvert PSL(2,7)\rvert = \text{rank } E_8 \cdot b_2$, $b_3 + \dim G_2 = \dim K_7 \cdot \alpha_{sum}$) are arithmetic facts about the chosen integers. Earlier versions called them

an "algebraic web" and a "Fibonacci–Lucas embedding" reflecting icosahedral geometry. Small integers satisfy many such identities; no physical consequence follows from these, and they are not evidence.

S2.8 Withdrawn readings

Earlier statement	Why it is withdrawn
"Complete derivations", "proof", "VERIFIED", "TOPOLOGICAL" as statuses of the relations	the arithmetic is exact; the assignment is not derived (S2.0)
$N_{gen} = 3$ by the Atiyah–Singer index theorem	no operator is specified; the expression is the integer identity of S2.1
"All 33 predictions use only topological invariants; no fitting"	eight use the imposed normalization or κ_T (R2, R3); the formulas were chosen with the values known (R4)
Mean deviation 0.73%, "11 exact matches", deviation histograms	computed against an earlier dataset; a mean deviation measures closeness, which is not distinctive (main §4.2, §7)
Statistical uniqueness: 192 349 alternatives, none better, $P < 5 \times 10^{-6}$, gauge-group and holonomy rankings	the alternatives were scored with the framework's own formulas, not allowed to search; replaced by the preregistered null of main §7 and S4
"Literature uniqueness" of (21, 77)	absence from a catalogue is not selection; the selection audit is main §10
Results "from analysis of the NK-certified metric": V_{min} formula, $U(1)^2$ symmetry, origin of the SM gauge group from triality	properties of an approximate seam metric (main §9); not used by any relation, and the gauge-group statement is not a derivation (main §2)
$\kappa_T = 1/61$ as a "topological torsion capacity"; $61 = \dim F_4 + {N_{gen}}^2$	an integer combination without a physical name (R3); a second expression for 61 is another instance of S2.7
$\det(g) = 65/32$ "consistent with the topological formula" via three algebraic paths	imposed normalization (R2); the metric optimization was constrained to this target
3477 "factorizes into framework constants"; associative-volume "cross-confirmation"	S2.3
$\Omega_m = \Omega_{DE}/\sqrt{5}$, Ω_{DM}/Ω_b "exact", ln 2 as an information factor	S2.6

Appendix F of earlier versions: the compactification factor 62/69. A residual search after the NuFIT 6.0 release found $197 \times 62/69 = 177.01$, matching that release's central value, and earlier versions kept it as a "contingency plan": if DUNE measured δ_{CP} near 177°, the framework would adopt 177.01 as its prediction and read 197 as a "bare" value. That plan is a repair clause, and version 3.6 has none (main §11, rule D1). It is withdrawn without condition: if a measurement excludes 197° at 3σ, the relation $\delta_{CP} =$

197 and its unit assignment are rejected, and no factor found afterwards replaces them. The factor 62/69 = 248/276 is recorded in S2.7 as one more formula the vocabulary offers.

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